

Multivariate descriptive statistics

Describing multivariate data sets

1 DATA STRUCTURE AND INTERPRETATION

2 MULTIVARIATE DESCRIPTIVE MEASURES

- Mean vector, covariance matrix, correlation matrix. Properties.
- Eigenanalysis reveals linear dependencies in the observed variables
- Linear dependency measures

3 STATISTICAL DISTANCE

4 OUTLIERS

5 DATA TRANSFORMATIONS

6 PLOTS

OVERALL GOALS

- To present the general structure of a multivariate data set.
- To introduce several multivariate descriptive measures.
- Multivariate Outliers. Data transformations.
- To review some techniques for visualizing multivariate data.

REFERENCES

Visit *Bibliography and other resources* on Moodle.

-  Johnson, R.A., Whichern, D.W. (2007) Applied Multivariate Statistical Analysis. Pearson Education.
-  Rencher, A.C. Methods of Multivariate Analysis. (2002) Wiley-Interscience.
-  (In Spanish) Peña, D. Análisis de datos Multivariantes. McGraw Hill, 2002.
-  Everitt, B.S., Dunn G. Applied Multivariate Data Analysis. (1997) Arnold.

DATA MATRIX

We observe p variables $(X_1, X_2, \dots, X_p)$ on n *individuals, items or trials*.
We placed them in an $n \times p$ array (matrix):

$$X_{n \times p} = \begin{pmatrix} x_{11} & x_{12} & \cdots & x_{1p} \\ x_{21} & x_{22} & \cdots & x_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{np} \end{pmatrix} = [x_1, \dots, x_p] = \begin{pmatrix} x'_1 \\ \vdots \\ x'_n \end{pmatrix}$$

where x_{ij} is the measure on individual i -th and variable j -th, and:

$$i = 1, \dots, n \quad j = 1, \dots, p$$

DATA MATRIX. EXAMPLE

```
> hbat
      X1 X2 X3 X4 X5   X6  X7  X8  X9 X10 X11 X12 X13 X14 X15 X16 X17 X18 X19 X20 X21  X22 X23
1      2  0  1  1  1  8.5 3.9 2.5 5.9 4.8 4.9 6.0 6.8 4.7 4.3 5.0 5.1 3.7 8.2 8.0 8.4 65.1   1
2      3  1  0  0  0  8.2 2.7 5.1 7.2 3.4 7.9 3.1 5.3 5.5 4.0 3.9 4.3 4.9 5.7 6.5 7.5 67.1   0
3      3  0  1  1  1  9.2 3.4 5.6 5.6 5.4 7.4 5.8 4.5 6.2 4.6 5.4 4.0 4.5 8.9 8.4 9.0 72.1   1
4      1  1  1  1  0  6.4 3.3 7.0 3.7 4.7 4.7 4.5 8.8 7.0 3.6 4.3 4.1 3.0 4.8 6.0 7.2 40.1   0
5      2  0  1  0  1  9.0 3.4 5.2 4.6 2.2 6.0 4.5 6.8 6.1 4.5 4.5 3.5 3.5 7.1 6.6 9.0 57.1   0
6      1  1  0  1  0  6.5 2.8 3.1 4.1 4.0 4.3 3.7 8.5 5.1 9.5 3.6 4.7 3.3 4.7 6.3 6.1 50.1   0
7      1  1  1  1  0  6.9 3.7 5.0 2.6 2.1 2.3 5.4 8.9 4.8 2.5 2.1 4.2 2.0 5.7 7.8 7.2 41.1   0
8      2  0  1  1  0  6.2 3.3 3.9 4.8 4.6 3.6 5.1 6.9 5.4 4.8 4.3 6.3 3.7 6.3 5.8 7.7 56.1   0
9      2  1  1  1  0  5.8 3.6 5.1 6.7 3.7 5.9 5.8 9.3 5.9 4.4 4.4 6.1 4.6 7.0 7.5 8.2 56.1   1
10     1  0  1  1  0  6.4 4.5 5.1 6.1 4.7 5.7 5.7 8.4 5.4 5.3 4.1 5.8 4.4 5.5 5.9 6.7 59.1   0
11     3  0  1  0  1  8.7 3.2 4.6 4.8 2.7 6.8 4.6 6.8 5.8 7.5 3.8 3.7 4.0 7.4 7.0 8.4 68.1   0
.....
```

100 observations on 23 variables.

VARIABLES

Variable Description		Variable Type
<u>Data Warehouse Classification Variables</u>		
X1	Customer Type	nonmetric
X2	Industry Type	nonmetric
X3	Firm Size	nonmetric
X4	Region	nonmetric
X5	Distribution System	nonmetric
<u>Performance Perceptions Variables</u>		
X6	Product Quality	metric
X7	E-Commerce Activities/Website	metric
X8	Technical Support	metric
X9	Complaint Resolution	metric
X10	Advertising	metric
X11	Product Line	metric
X12	Salesforce Image	metric
X13	Competitive Pricing	metric
X14	Warranty & Claims	metric
X15	New Products	metric
X16	Ordering & Billing	metric
X17	Price Flexibility	metric
X18	Delivery Speed	metric
<u>Outcome/Relationship Measures</u>		
X19	Satisfaction	metric
X20	Likelihood of Recommendation	metric
X21	Likelihood of Future Purchase	metric
X22	Current Purchase/Usage Level	metric
X23	Consider Strategic Alliance/Partnership in Future	nonmetric

DATA MATRIX. EXAMPLE

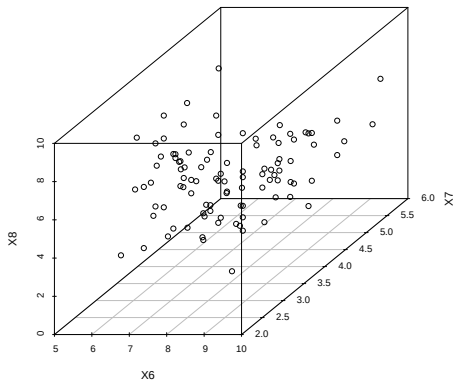
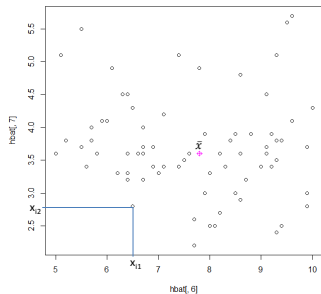
```
> medifis
  V1  V2 V3  V4  V5  V6  V7  V8
1   0 159 49 36.0 68.0 42.0 57.0 40.0
2   1 164 62 39.0 73.0 44.0 55.0 44.0
3   0 172 65 38.0 75.0 48.0 58.0 44.0
4   0 167 52 37.0 73.0 41.5 58.0 44.0
5   0 164 51 36.0 71.0 44.5 54.0 40.0
6   0 161 67 38.0 71.0 44.0 56.0 42.0
7   0 168 48 39.0 72.5 41.0 54.5 43.0
8   1 181 74 43.0 74.0 50.0 60.0 47.0
9   1 183 74 41.0 79.0 47.5 59.5 47.0
10  0 158 50 36.0 68.5 44.0 57.0 41.0
11  0 156 65 36.0 68.0 46.0 58.0 41.0
12  1 173 64 40.0 79.0 48.0 56.5 47.0
13  0 158 43 36.0 68.0 43.0 55.0 39.0
14  1 178 74 42.0 75.0 50.0 59.0 45.0
15  1 181 76 43.0 83.0 51.0 57.0 43.0
16  1 182 91 41.0 83.0 53.0 59.0 43.0
17  1 176 73 42.0 78.0 48.0 58.0 45.0
18  0 162 68 39.0 72.0 44.0 59.0 42.0
.....
```

- V1 =gender, V2 =height
- V3 =weight, V4 =foot length
- V5 =arm length
- V6 =back width
- V7 =skull diameter
- V8 =knee-ankle length

27 observations on 8 variables.

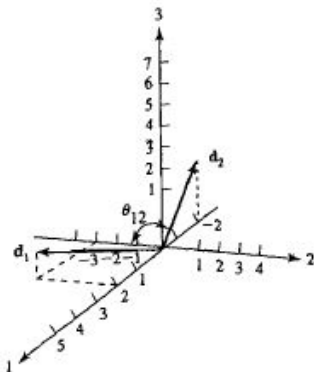
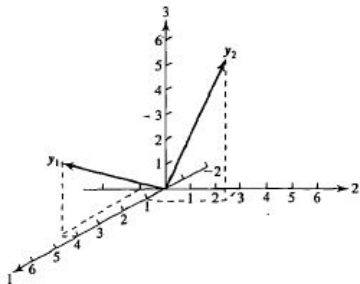
INTERPRETATION I

n points in p -dimensional space. Location and variability of the points.



INTERPRETATION II

p vectors in n -dimensional space. Representing columns of the matrix.



MULTIVARIATE DESCRIPTIVE MEASURES

Mean vector and covariance matrix.

MEAN VECTOR

p -dimensional vector containing averages for every variable.

$$\bar{\mathbf{x}} = \frac{1}{n} \sum_{i=1}^n \mathbf{x}_i = \begin{bmatrix} \bar{x}_1 \\ \vdots \\ \bar{x}_p \end{bmatrix}$$

COVARIANCE MATRIX

$$S_{p \times p} = \frac{1}{n} \sum_{i=1}^n (\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})' = \begin{pmatrix} s_1^2 & s_{12} & \cdots & s_{1p} \\ s_{12} & s_2^2 & \cdots & s_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ s_{p1} & s_{p2} & \cdots & s_p^2 \end{pmatrix}$$

MULTIVARIATE DESCRIPTIVE MEASURES

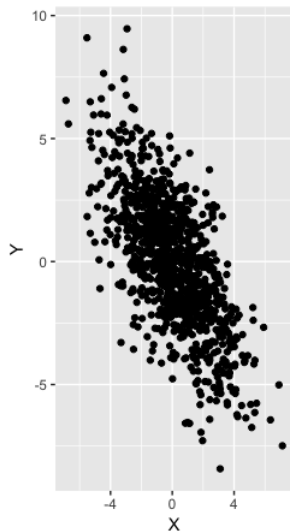
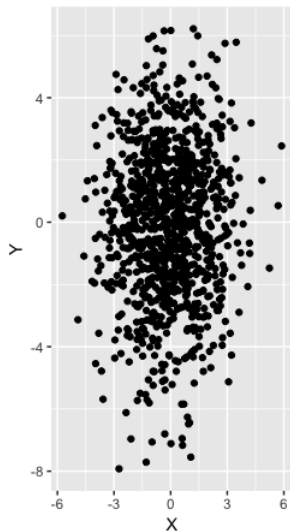
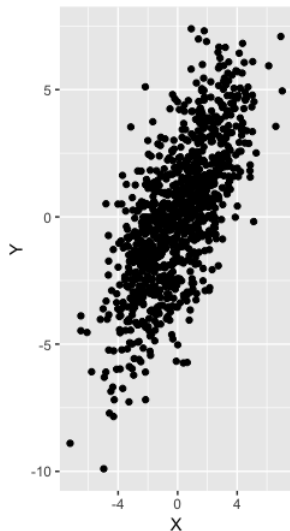
Covariance Matrix:

- It contains the variance of each variable in the diagonal and off-diagonal entries are the covariances between any two variables.
- Variance and covariance depend on the measurement units of the variables. 😞 😞
- The covariance *measures* the linear dependency between the observations of each pair of variables.

The covariance matrix contains all the information about the spread (variances) and orientation (covariance, linear dependence between each pair of variables) of our data.

MULTIVARIATE DESCRIPTIVE MEASURES

The covariance matrix defines the overall shape of the data.



MEAN VECTOR. EXAMPLE.

```
> colMeans(hbat)
  X1    X2    X3    X4    X5    X6    X7    X8
2.010 0.480 0.510 0.610 0.430 7.810 3.672 5.365
  X9   X10   X11   X12   X13   X14   X15   X16
5.442 4.010 5.805 5.123 6.974 6.043 5.150 4.278
  X17   X18   X19   X20   X21   X22   X23
4.610 3.886 6.918 7.020 7.713 58.400 0.450

> colMeans(medifis)
  V1    V2    V3    V4    V5
0.4444444 168.7777778 63.8888889 38.9814815 73.4629630
  V6    V7    V8
45.8518519 57.2407407 43.0925926
```

Mean vector: multivariate center of the data set. You can look for disparities in the values of the mean vector components. Check, for instance, the dataset *state.x77* in R.

S MATRIX. EXAMPLE.

```
> cov(medifis[,-1])
```

	V2	V3	V4	V5	V6	V7	V8
V2	103.94872	108.35897	27.091880	45.856838	34.427350	11.036325	27.155983
V3	108.35897	163.87179	31.151709	52.053419	43.213675	14.604701	29.029915
V4	27.09188	31.15171	8.201567	12.095442	9.189459	2.889245	7.694088
V5	45.85684	52.05342	12.095442	24.575499	15.975071	4.336182	11.974715
V6	34.42735	43.21368	9.189459	15.975071	16.169516	4.642806	8.004630
V7	11.03632	14.60470	2.889245	4.336182	4.642806	3.391738	3.226852
V8	27.15598	29.02991	7.694088	11.974715	8.004630	3.226852	9.962251

```
> det(cov(medifis[,-1]))
```

```
[1] 277425.3
```

```
> det(cov(medifis[,-1]))^(1/7)
```

```
[1] 5.992283
```

```
> det(cov(medifis[,-1]))^(1/14)
```

```
[1] 2.447914
```

S MATRIX. PROPERTIES.

- Symmetric, positive semi-definite (or nonnegative definite).
- Non negative trace, determinant and eigenvalues.
- Its rank determines the number of linearly independent variables in the data matrix and it is also the number of eigenvalues different from 0.

DIMENSIONALITY THEOREM

Being $r = \text{rank}(S) \leq p$ there are r linearly independent variables in the data set. The others $p - r$ are linear combinations of these r variables.

GLOBAL VARIABILITY MEASURES

GENERALIZED VARIANCE

$|S|$. It has intuitively pleasant geometrical interpretations.

EFFECTIVE VARIANCE (PEÑA AND RODRÍGUEZ, 2003)

$|S|^{1/p}$.

TOTAL VARIANCE

It's the trace of S . $T = tr(S) = \sum_{i=1}^p s_i^2 = \sum_{i=1}^p \lambda_i$.

MEAN VARIANCE

$\bar{s}^2 = \frac{1}{p} \sum_{i=1}^p s_i^2$.

GENERALIZED VARIANCE DISADVANTAGES

Scatterplots with different orientations but the same value for $|S|$.

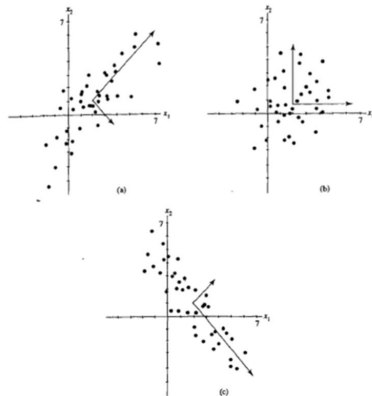


Figure 3.8 from Johnson, Wichern (2007)

EQUIVALENT CONDITIONS FOR $|S| = 0$

When $|S| = 0$, there are some variables that are linear combinations of the others.

Whenever a non-zero vector $\mathbf{a}$ satisfies one of the following conditions, it satisfies all of them:

- 1 $\mathbf{S}\mathbf{a} = \mathbf{0}$, $\mathbf{a}$ is a scaled eigenvector of S with eigenvalue 0. The eigenvector $\mathbf{a}$ gives coefficients for the linear dependency of the mean corrected data.
- 2 $\mathbf{a}'(\mathbf{x}_j - \bar{\mathbf{x}}) = 0$ The linear combination of the mean corrected data, using $\mathbf{a}$, is 0.
- 3 $\mathbf{a}'\mathbf{x}_j = c, \forall j (c = \mathbf{a}'\bar{\mathbf{x}})$ The linear combination of the original data, using $\mathbf{a}$, is constant.

EXAMPLE 1. $|S| = 0$

Let S_1 be a variance-covariance matrix of a random sample on 4 variables:

```
> ese1
      var1   var2   var3   var4
1 0.0947 0.0242 0.0054 0.0594
2 0.0242 0.0740 0.0285 0.0491
3 0.0054 0.0285 0.0838 0.0170
4 0.0594 0.0491 0.0170 0.0543

> det(ese1)
[1] 3.496301e-08

> eigen(ese1)
$values
[1] 1.729668e-01 8.761555e-02 4.616766e-02 4.997203e-05

$vectors
      [,1]      [,2]      [,3]      [,4]
[1,] -0.5937114 0.5270237 0.4509023 -0.407970466
[2,] -0.5046316 -0.2794621 -0.7073918 -0.408466309
[3,] -0.3019646 -0.7930913 0.5289833 -0.000437213
[4,] -0.5492459 0.1230977 -0.1282992 0.816526290
```

EXAMPLE 2. $|S| = 0$

Let S_2 be a variance covariance matrix with eigenvalues:

0.0019, 0.0649, 0.4208, 1.046, 2.4434, 5.6394, 15.207, 43.7017, 303.458

- First eigenvalue is very close to 0. Its eigenvector, $(.335, .324, .337, .339, .325, .337, .334, .332, .334)$, has got very similar components, showing that the sum of the variables is almost constant.
- Second eigenvalue is also very close to 0. Its eigenvector is $(-0.07, -0.29, -0.07, 0.91, 0.06, -0.1, -0.12, -0.05, -0.22)$. Assuming all values are close to 0 except the fourth 0.91, we can conclude that this variable has very small variance, it is almost constant through the observed data.

IMPLICATIONS OF SINGULARITY OR NEAR-SINGULARITY OF S ($|S| = 0$)

Some frequent specific situations would be:

- Two or more variables sum up to a constant.
- Two variables are identical or differ merely in the mean or variance.
- Creating new variables that are sums of the original ones.

LINEAR DEPENDENCY MEASURES

- Pairwise linear dependence: R matrix.
- Coefficient of determination: squared multiple correlation coefficient R^2 .
- Partial correlations (pairwise).
- Overall dependence measure: effective dependence.

PAIRWISE LINEAR DEPENDENCE: R MATRIX

Covariances are difficult to interpret, so often it is useful to work with the correlation, which is always between -1 and 1 .

R is the correlation matrix: square, symmetric, filled with 1 's in the principal diagonal and with pairwise Pearson's correlation coefficients elsewhere.

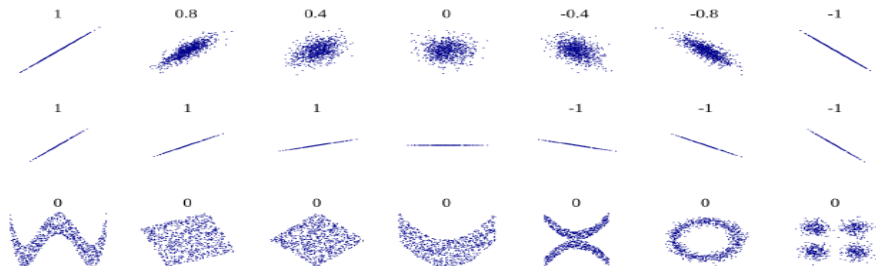
$$R = \begin{pmatrix} 1 & r_{12} & \cdots & r_{1p} \\ r_{21} & 1 & \cdots & r_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ r_{p1} & r_{p2} & \cdots & 1 \end{pmatrix}$$

```
> cor(medifis[,-1])
```

	V2	V3	V4	V5	V6	V7	V8
V2	1.0000000	0.8302392	0.9278572	0.9072842	0.8397413	0.5877651	0.8438733
V3	0.8302392	1.0000000	0.8497308	0.8202495	0.8394998	0.6194830	0.7184804
V4	0.9278572	0.8497308	1.0000000	0.8519658	0.7979818	0.5478035	0.8511969
V5	0.9072842	0.8202495	0.8519658	1.0000000	0.8013885	0.4749471	0.7653063
V6	0.8397413	0.8394998	0.7979818	0.8013885	1.0000000	0.6269327	0.6306872
V7	0.5877651	0.6194830	0.5478035	0.4749471	0.6269327	1.0000000	0.5551229
V8	0.8438733	0.7184804	0.8511969	0.7653063	0.6306872	0.5551229	1.0000000

DIFFERENT BIVARIATE PATTERNS AND THEIR CORRELATION

- Geometrically, you can think of the correlation between two variables as the cosine of the angle between two vectors.



COEFFICIENT OF DETERMINATION

- Squared multiple correlation coefficient R^2 between a variable X_j and a set of (*independent, predictors*) variables $(X_1, \dots, X_{j-1}, X_{j+1}, \dots, X_p)$ is defined in the context of multiple linear regression. It measures the *goodness* of a linear combination of the independent variables to predict X_j .

$$R_{j.1\dots p}^2$$

- It is calculated from the diagonal elements of S and S^{-1} .

```
> 1-1/(diag(solve(cov(medifis[, -1])))*diag(cov(medifis[, -1])))  
      V2      V3      V4      V5      V6      V7      V8  
0.9303971 0.8195433 0.8998983 0.8540904 0.8132550 0.5188673 0.7897407
```

Height (V2) is the best *linearly explained by the others* ($R^2 = 0.93$), followed by foot length (V4, $R^2 = 0.89$) and arm length (V5, $R^2 = 0.85$). The worst *linearly explained by the others* is skull diameter (V7, $R^2 = 0.51$).

PARTIAL CORRELATIONS

- Partial correlation between variables (X_1, X_2) , controlling for $(X_3, \dots, X_p)$, $r_{12.3\dots p}$, takes into account (removes) the external influences of the rest of the variables.
- For instance, P matrix for 4 variables would be:

$$P = \begin{pmatrix} 1 & r_{12.34} & r_{13.24} & r_{14.23} \\ r_{21.34} & 1 & r_{23.14} & r_{24.13} \\ r_{31.24} & r_{32.14} & 1 & r_{34.12} \\ r_{41.23} & r_{42.13} & r_{43.12} & 1 \end{pmatrix}$$

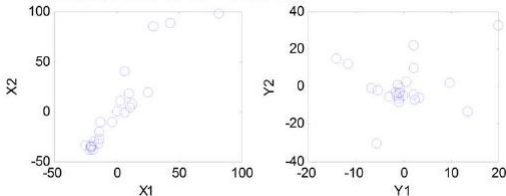
where, for example, $r_{12.34}$ is the partial correlation of (X_1, X_2) controlling for X_3 y X_4 .

PARTIAL CORRELATIONS. EXAMPLE 1.



$$R = \begin{pmatrix} 1 & 0.9331 & 0.9599 \\ 0.9331 & 1 & 0.9553 \\ 0.9599 & 0.9553 & 1 \end{pmatrix}$$

$$r_{12.3} = 0.1943 \quad (p\text{-}v = 0.363)$$



PARTIAL CORRELATIONS. EXAMPLE 2.

```
> cor2pcor(cor(medifis[-1]))
```

	[,1]	[,2]	[,3]	[,4]	[,5]	[,6]	[,7]
[1,]	1.0000000	-0.190451632	0.48165069	0.51584105	0.31943604	0.1646827	0.272114917
[2,]	-0.1904516	1.000000000	0.37047271	0.29497085	0.34123019	0.2607606	0.002812968
[3,]	0.4816507	0.370472712	1.00000000	-0.10729218	0.07020131	-0.2037158	0.366228245
[4,]	0.5158410	0.294970855	-0.10729218	1.00000000	0.12778058	-0.3124446	0.094853134
[5,]	0.3194360	0.341230186	0.07020131	0.12778058	1.00000000	0.2861702	-0.372949025
[6,]	0.1646827	0.260760594	-0.20371581	-0.31244462	0.28617015	1.0000000	0.270044241
[7,]	0.2721149	0.002812968	0.36622824	0.09485313	-0.37294903	0.2700442	1.000000000

Stronger partial relationships are between $V5[,4]$ and $V2[,1]$ (arm length and height, 0.515) and between $V4[,3]$ and $V2[,1]$ (foot length and height, 0.48).

PARTIAL CORRELATIONS. EXAMPLE 3

- Sample of $n = 12$ young men, aged 12 to 19.
- Correlation between height (X_1) and weight (X_2) will be high because older $\approx$ taller on average and older $\approx$ heavier on average. So, we recognize the presence of a third variable age (X_3).
- We want to know if there is a real correlation between height (X_1) and weight (X_2):
 - Correlation coefficient between height and weight $r_{12} = 0.78$
 - Correlation coefficient between height and age $r_{13} = 0.52$
 - Correlation coefficient between weight and age $r_{23} = 0.54$
 - Partial correlation $r_{12.3} = 0.69$. It is significant at 0.2% level (2 chances in 1000 that it could arise by chance if NO true correlation were present)

$$r_{12.3} = \frac{r_{12} - r_{13}r_{23}}{\sqrt{(1 - r_{13}^2)(1 - r_{23}^2)}}$$

¡Careful, the formula changes for $r_{12.34}$!

MULTIVARIATE LINEAR DEPENDENCE: EFFECTIVE DEPENDENCE

Properties of $|\mathbf{R}|$:

- $0 \leq |\mathbf{R}| \leq 1$. If the variables are pairwise uncorrelated, $\mathbf{R}$ is a diagonal matrix of 1's and $|\mathbf{R}| = 1$.
- If one variable is a linear combination of the others, $\mathbf{S}$ and $\mathbf{R}$ are singular matrices and $|\mathbf{R}| = 0$.
- The following formula holds:

$$|\mathbf{R}_p| = (1 - R_{p.1\dots p-1}^2) \cdot (1 - R_{p-1.1\dots p-2}^2) \cdot \dots \cdot (1 - R_{2.1}^2)$$

EFFECTIVE DEPENDENCE COEFFICIENT

$$D(\mathbf{R}_p) = 1 - |\mathbf{R}_p|^{1/(p-1)}$$

It is a multivariate overall measure of linear dependence. It represents the average proportion of explained variability among the variables due to linear dependencies.

EFFECTIVE DEPENDENCE EXAMPLE

```
> 1-det(cor(medifis[-1]))^(1/6)
[1] 0.7700371
```

Altogether, linear dependences explain 77% of the variability in the data set.

DISTANCE

To obtain a useful distance measurement in a multivariate setting, we must consider not only variances but also their covariances or correlations.

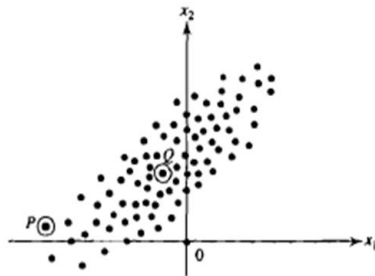
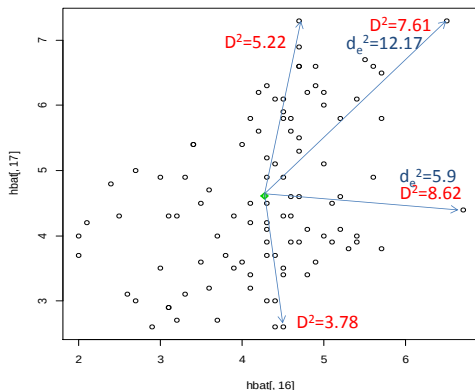


Figure 1.25 from Johnson, Wichern (2007)

DISTANCE

Mahalanobis distance in red, euclidean in blue. Correlation coefficient = 0.4.



```
> cov(cbind(hbat[,16], hbat[,17]))  
      [,1]      [,2]  
[1,] 0.8627434 0.4558788  
[2,] 0.4558788 1.4544444  
> cor(cbind(hbat[,16], hbat[,17]))  
      [,1]      [,2]  
[1,] 1.0000000 0.4069677  
[2,] 0.4069677 1.0000000
```

DISTANCE

MAHALANOBIS DISTANCE

The (squared) distance between two vectors is defined as:

$$D_j = [(\mathbf{x}_j - \bar{\mathbf{x}})' S^{-1} (\mathbf{x}_j - \bar{\mathbf{x}})]^{(1/2)}$$

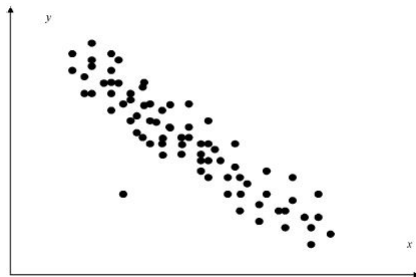
or

$$D_j^2 = [(\mathbf{x}_j - \bar{\mathbf{x}})' S^{-1} (\mathbf{x}_j - \bar{\mathbf{x}})]$$

This distance can be used to detect outliers. If the data comes from a p -dimensional Normal variate its distribution follows a χ_p^2 .

OUTLIERS IDENTIFICATION

- Univariate (using, for example, the boxplot)
- Multivariate, using Mahalanobis distance or robust estimators.

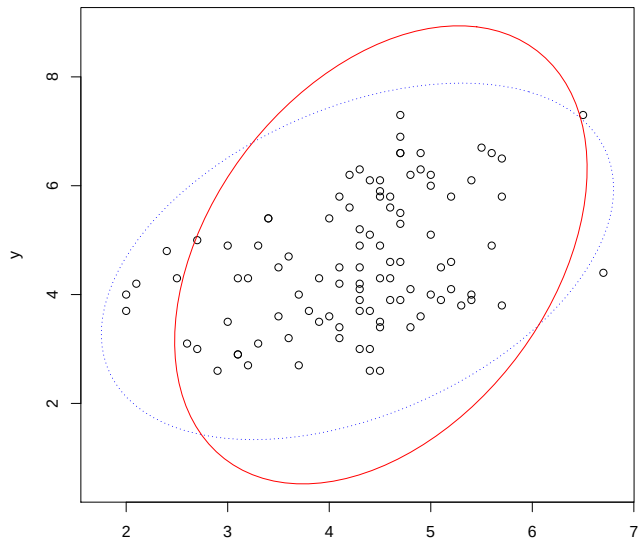


BIVARIATE OUTLIERS

```
> corr.plot(hbat[,16], hbat[,17])
```

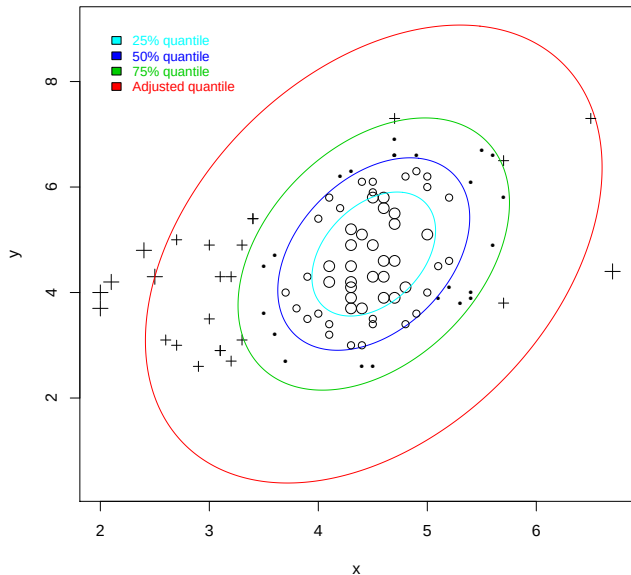
Classical cor = 0.41

Robust cor = 0.38



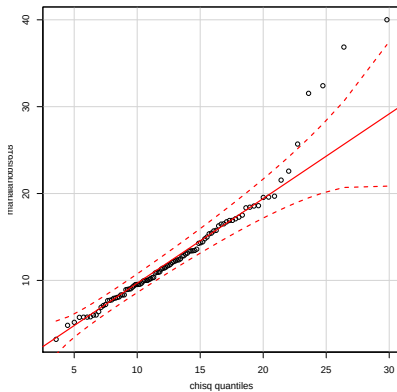
BIVARIATE OUTLIERS

```
> symbol.plot(cbind(hbat[,16], hbat[,17]))
```



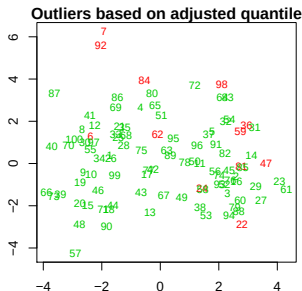
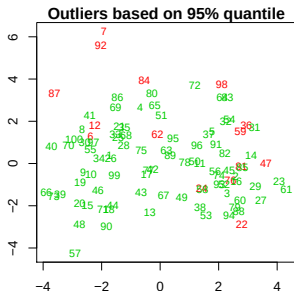
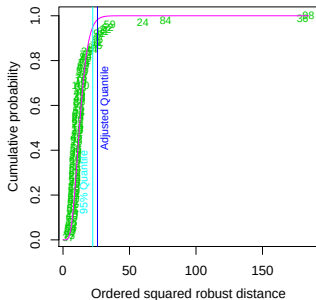
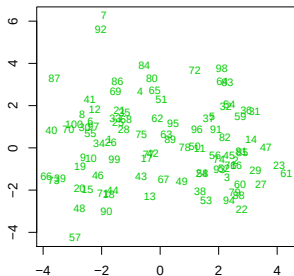
MULTIVARIATE OUTLIERS

- Method based on the Mahalanobis distance (by hand, with *R*).
 - 1 Compute, for each observation, the Mahalanobis distance to the mean vector, d_i^2 .
 - 2 Examine these distances for unusually large values. In a chi-squared plot, these would be the points farthest from the origin. Look for values that exceed the upper fifth percentile of the chi-squared distribution.
- Robust method based on Mahalanobis distance (package *mvoutlier* of *R*).



MULTIVARIATE OUTLIERS

```
> aq.plot(hbat[,6:18])
```



LINEAR TRANSFORMATIONS

Two important linear transformations that are often used with multivariate data are:

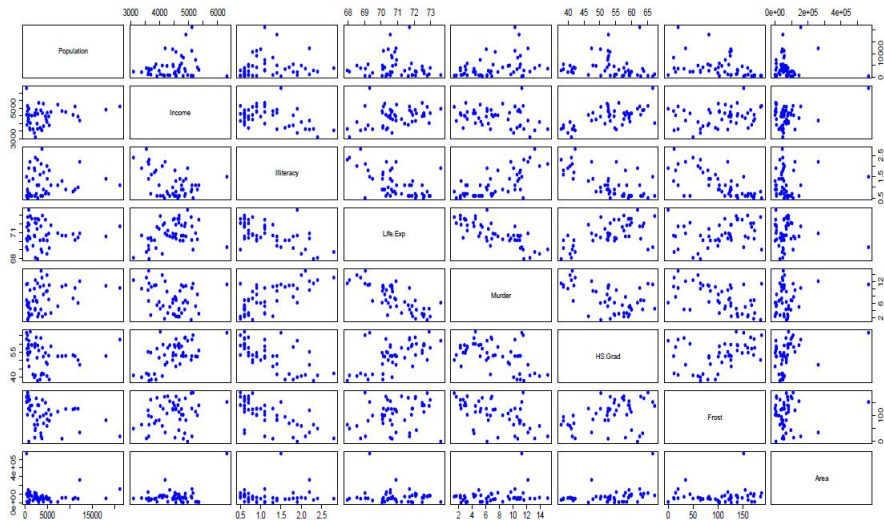
- Individual standardization of each variable: eliminates the mean and standardises the variance of each variable.
- Multivariate standardization: eliminates the mean and the correlation between the variables and standardises the variance of each one.

The next three scatterplot matrices correspond to:

- The original data set.
- The univariate standardised data set.
- The multivariate standardised data set.

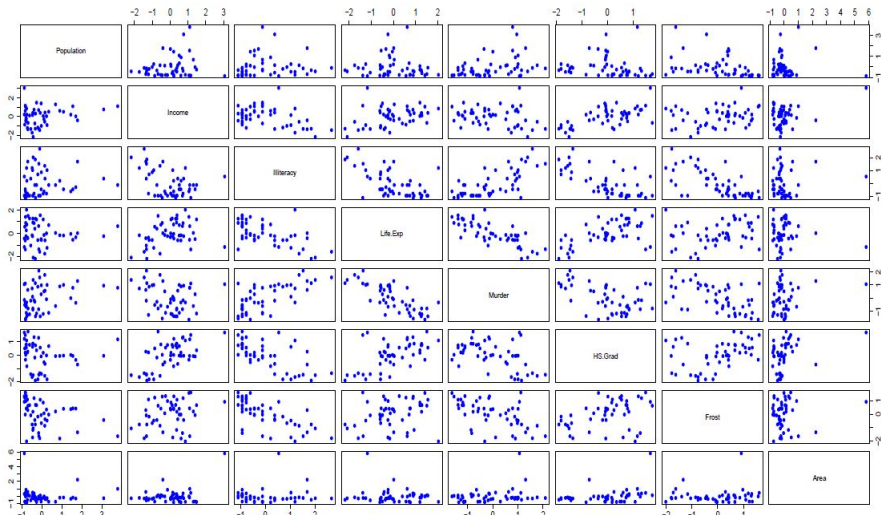
LINEAR TRANSFORMATIONS

Original Data set



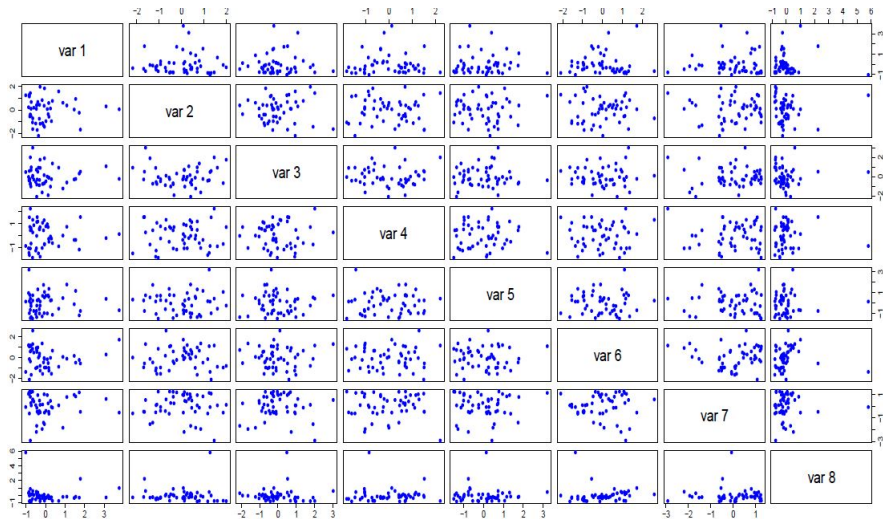
LINEAR TRANSFORMATIONS

Univariate standardised Data set



LINEAR TRANSFORMATIONS

Multiivariate standardised Data set



DATA TRANSFORMATIONS TO NEAR NORMALITY

Box-Cox transformation:

$$x^{(\lambda)} = \begin{cases} \frac{(x+m)^\lambda - 1}{\lambda} & (\lambda \neq 0) \quad (x > -m) \\ \ln(x+m) & (\lambda = 0) \quad (m > 0) \end{cases}$$

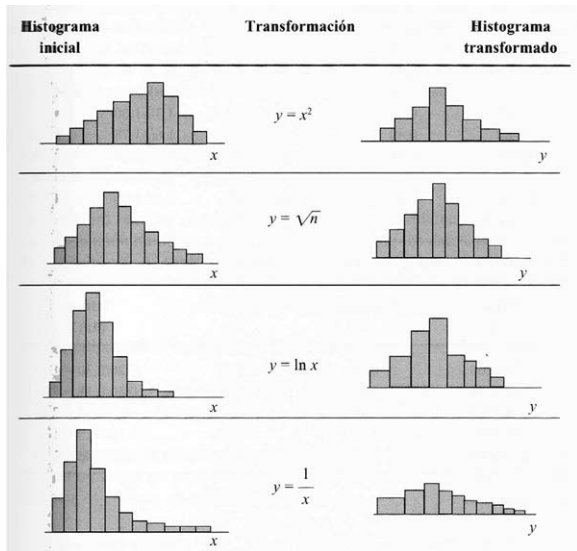
$$x^{-1} = \frac{1}{x}, \quad x^0 = \ln x, \quad x^{1/4} = \sqrt[4]{x}, \quad x^{1/2} = \sqrt{x} \quad \dots \dots$$

Shrinks large values of x ...

$$\dots \dots x^2, \quad x^3$$

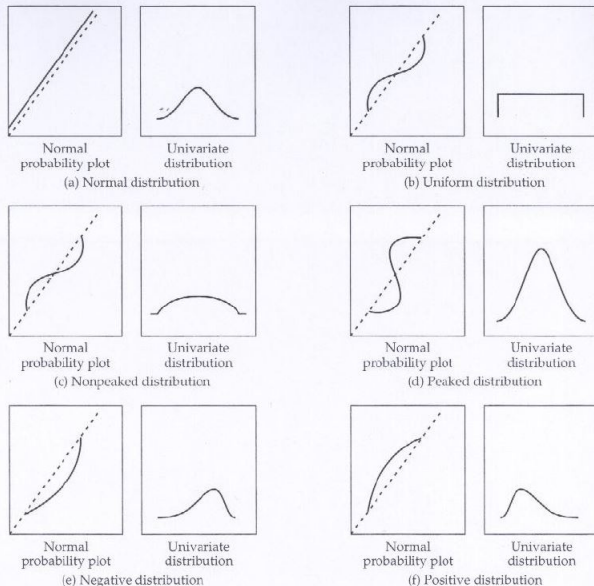
... increases large values of x

DATA TRANSFORMATIONS



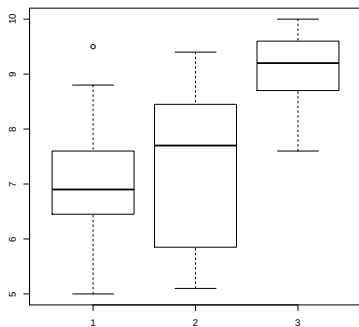
NORMAL PROBABILITY PLOTS

Normal probability plots and corresponding univariate distributions

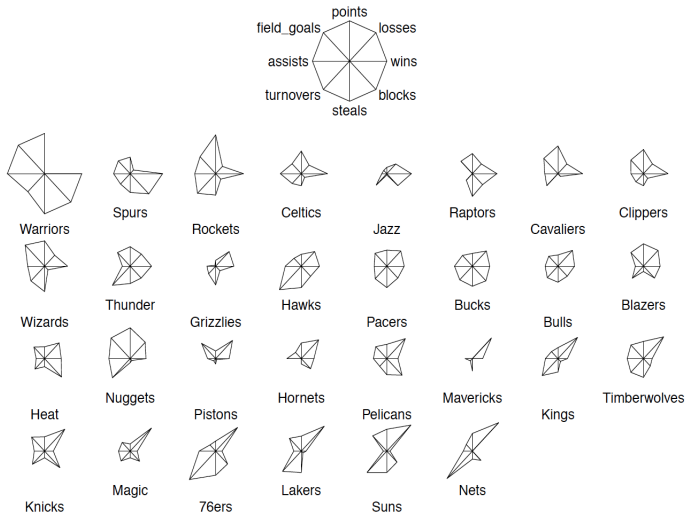


PLOTS

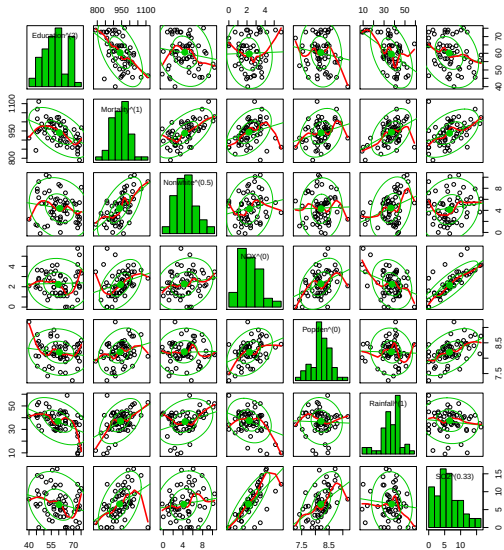
- Univariate Plots
- Multivariate plots (Matrix Scatterplot, bubble plots, Coplots, Andrews curves, contour plots...)
- Multivariate plots: Glyphplots (Chernoff faces, star plots, ...)



GLYPH PLOTS: AN EXAMPLE



PLOTS



ANDREWS' CURVES

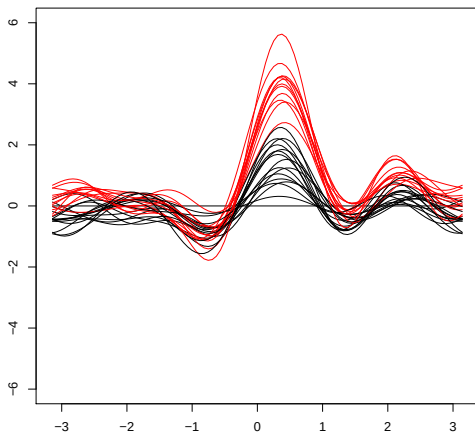
Each multivariate observation, $(x_1, \dots, x_p)$, is transformed into a curve:

$$x(t) = \frac{x_1}{\sqrt{2}} + x_2 \sin(t) + x_3 \cos(t) + x_4 \sin(2t) + x_5 \cos(2t) + \dots$$

This function is plotted over the range $-\pi \leq t \leq \pi$. A set of multivariate observations will appear as a set of lines on the plot.

- The distance between curves reflects correctly the pairwise distances between observations.
- Outliers appear as single Andrews' curves that look different from the rest.
- A subgroup of data is characterized by a set of similar curves.
- The order of variables does affect the appearance of the plot.

ANDREWS' CURVES



GOALS OF MULTIVARIATE ANALYSIS

- Some multivariate analyses are *exploratory*, in which the investigator is merely seeking patterns in the data and looking for compelling behaviour.
- Many exploratory methods involve data summary, data reduction and graphics.
- Other multivariate data analysis are confirmatory, in which the investigator has well-defined hypothesis he or she wishes to test.
- Here, inference such as significance test can be useful.
- Many confirmatory analyses have specific conditions required for the conclusions to be valid.